

Company-Wide KPI Scorecard Dashboard

Aurelia Financial Services S.A. | Power BI Portfolio Project

Project Overview

Dashboard link: This document is the supporting project record for the Aurelia KPI Scorecard. It is not the primary dashboard deliverable; the core deliverables are the Power BI report file, the published dashboard link, and a clean PDF export. This companion document records the business objective, model design, implementation choices, and measure logic used to build the report.

Role	Business Intelligence Analyst
Organisation	Aurelia Financial Services S.A. (fictional)
Reporting period	January–December 2025. Prior-year measures are included in the model design but return blank because the source dataset only contains 2025 data.
Markets	Luxembourg, Germany, France, and Belgium
Technology	Power BI Desktop, Power Query, DAX, and Excel

Business Objective

Build a senior-management scorecard covering Revenue, Cost, Customer, and Operations, with interactive filtering, RAG status, a revenue drill-through page, and an operations deep-dive page.

Portfolio Deliverables

- Power BI report file (.pbix)
- Published Power BI Service dashboard link
- PDF export or clean dashboard screenshots
- This supporting build and methodology document

Published Dashboard

[Aurelia Financial Services KPI Dashboard](#) THE DASHBOARD LINK

Implementation Record

Source Workbook and Model

The source workbook is **Aurelia_KPI_Data.xlsx**. It contains Fact_Performance, Dim_Date, Dim_Region, Dim_Product, Dim_Department, and KPI_Targets. The fact table grain is one row per Month × Region × Product, producing 192 rows for 2025.

The model uses one-to-many, single-direction relationships from the dimension tables to Fact_Performance. A contiguous daily calendar was created in Power BI and related to Fact_Performance by the Date column. The original 12-row monthly date sheet remains source metadata but is not suitable for marking as a date table.

Data modelling limitation: The source workbook contains a mixed-grain design. Financial metrics (Revenue, Cost, Transactions, AUM and loan balances) are correctly stored at Month × Region × Product grain. Customer and operational metrics (Active Customers, New Customers, Churned Customers, Complaints, Tickets, SLA activity and FTE) are recorded at Month × Region grain and repeated across product rows. In a production implementation, these measures would ideally be moved into a separate operational fact table related to the same dimensions. For this portfolio project, the issue was corrected through DAX aggregation logic.

Power Query Transformation

The following transformation sets explicit data types for the fact table:

```
= Table.TransformColumnTypes("#Promoted Headers",{{"Revenue_EUR", Currency.Type}, {"Revenue_Budget_EUR", Currency.Type}, {"Operating_Cost_EUR", Currency.Type}, {"Cost_Budget_EUR", Currency.Type}, {"Active_Customers", Int64.Type}, {"Active_Customer_Target", Int64.Type}, {"New_Customers", Int64.Type}, {"Churned_Customers", Int64.Type}, {"AUM_EUR", Currency.Type}, {"AUM_Target_EUR", Currency.Type}, {"Transaction_Count", Int64.Type}, {"Transaction_Target", Int64.Type}, {"Digital_Transaction_Count", Int64.Type}, {"Complaints", Int64.Type}, {"Total_Tickets", Int64.Type}, {"Tickets_Within_SLA", Int64.Type}, {"Gross_Loan_Balance_EUR", Currency.Type}, {"NPL_Balance_EUR", Currency.Type}, {"FTE", Int64.Type}, {"DateKey", Int64.Type}, {"Date", type date}, {"RegionKey", type text}, {"ProductKey", type text}, {"DepartmentKey", type text}})
```

Calendar Table

A daily calendar was created in Power BI so the Date table can be marked correctly and support time-intelligence functions.

Dim_Date =

```
ADDCOLUMNS(  
    CALENDAR(DATE(2025,1,1), DATE(2025,12,31)),  
    "DateKey", VALUE(FORMAT([Date], "YYYYMM")),  
    "Year", YEAR([Date]),  
    "Quarter", "Q" & FORMAT([Date], "Q"),  
    "MonthNumber", MONTH([Date]),  
    "MonthName", FORMAT([Date], "MMMM"),  
    "YearMonth", FORMAT([Date], "YYYY-MM")  
)
```

KPI Measure Library

The measures below implement the four KPI pillars and their RAG logic. Prior-year and year-over-year measures require 2024 fact data to return populated comparisons; with the current 2025-only workbook, those measures will be blank.

Semi-additive KPI correction note:

Several operational metrics (Active Customers, New Customers, Churned Customers, Complaints, Tickets, Tickets Within SLA, and FTE) are stored at Month × Region grain within a fact table whose physical grain is Month × Region × Product. To prevent duplication across product rows, DAX measures aggregate these fields at Date × Region level using SUMMARIZE and MAX before performing business calculations. Snapshot KPIs such as Active Customers and FTE are evaluated using closing-month logic.

REVENUE PILLAR

Total Revenue = SUM(Fact_Performance[Revenue_EUR])

Total Revenue Budget = SUM(Fact_Performance[Revenue_Budget_EUR])

Revenue YTD = TOTALYTD([Total Revenue], Dim_Date[Date])

Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))

Revenue vs Budget % = DIVIDE([Total Revenue], [Total Revenue Budget])

Revenue YoY % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY])

Revenue RAG =

```
SWITCH(TRUE(),  
  [Revenue vs Budget %] >= 1.00, " ● ",  
  [Revenue vs Budget %] >= 0.95, " ● ",  
  " ● "  
)
```

Revenue Growth YoY % = [Revenue YoY %] -- reuse above

Revenue Growth RAG =

```
SWITCH(TRUE(),  
  [Revenue Growth YoY %] >= 0.06, " ● ",  
  [Revenue Growth YoY %] >= 0.02, " ● ",  
  " ● "  
)
```

EBITDA = [Total Revenue] - [Operating Cost]

EBITDA Margin % = DIVIDE([EBITDA], [Total Revenue])

EBITDA Margin RAG =

```
SWITCH(TRUE(),  
  [EBITDA Margin %] >= 0.48, " ● ",
```

```

[EBITDA Margin %] >= 0.42, " ● ",
" ● "
)

```

AUM = CALCULATE(SUM(Fact_Performance[AUM_EUR]), LASTDATE(Dim_Date[Date]))

AUM Target = CALCULATE(SUM(Fact_Performance[AUM_Target_EUR]),
LASTDATE(Dim_Date[Date]))

AUM vs Target % = DIVIDE([AUM], [AUM Target])

AUM RAG =

```

SWITCH(TRUE(),
    [AUM vs Target %] >= 1.00, " ● ",
    [AUM vs Target %] >= 0.97, " ● ",
    " ● "
)

```

COST PILLAR

Operating Cost = SUM(Fact_Performance[Operating_Cost_EUR])

Operating Cost Budget = SUM(Fact_Performance[Cost_Budget_EUR])

Cost vs Budget % = DIVIDE([Operating Cost], [Operating Cost Budget])

Cost RAG = -- lower is better

```

SWITCH(TRUE(),
    [Cost vs Budget %] <= 1.00, " ● ",
    [Cost vs Budget %] <= 1.03, " ● ",
    " ● "
)

```

)

Cost-to-Income % = DIVIDE([Operating Cost], [Total Revenue])

Cost-to-Income PY = CALCULATE([Cost-to-Income %],
SAMEPERIODLASTYEAR(Dim_Date[Date]))

CIR RAG =

SWITCH(TRUE(),
[Cost-to-Income %] <= 0.48, " ,
[Cost-to-Income %] <= 0.53, " ,
" ,
)

Total FTE =

SUMX(
SUMMARIZE(
Fact_Performance,
Fact_Performance[Date],
Fact_Performance[RegionKey],
"RegionMonthFTE",
MAX(Fact_Performance[FTE])
),
[RegionMonthFTE]
)

Total FTE Closing =

CALCULATE(
[Total FTE],
LASTDATE(Dim_Date[Date])
)

Revenue per FTE =

DIVIDE(
[Total Revenue],

```

    [Total FTE Closing]
)

Revenue per FTE RAG =

    SWITCH(TRUE(),
        [Revenue per FTE] >= 180000, " ,
        [Revenue per FTE] >= 160000, " ,
        " ,
    )

```

CUSTOMER PILLAR

```

Active Customers =
SUMX(
    SUMMARIZE(
        Fact_Performance,
        Fact_Performance[Date],
        Fact_Performance[RegionKey],
        "RegionMonthActiveCustomers",
        MAX(Fact_Performance[Active_Customers])
    ),
    [RegionMonthActiveCustomers]
)

```

```

Active Customers Closing =
CALCULATE(
    [Active Customers],
    LASTDATE(Dim_Date[Date])
)

```

```

Active Customer Target =
SUMX(

```

```

SUMMARIZE(
    Fact_Performance,
    Fact_Performance[Date],
    Fact_Performance[RegionKey],
    "RegionMonthActiveCustomerTarget",
    MAX(Fact_Performance[Active_Customer_Target])
),
[RegionMonthActiveCustomerTarget]
)

```

Active Customer Target Closing =

```

CALCULATE(
    [Active Customer Target],
    LASTDATE(Dim_Date[Date])
)

```

Active Customers vs Target % =

```

DIVIDE(
    [Active Customers Closing],
    [Active Customer Target Closing]
)

```

Active Customers RAG =

```

SWITCH(
    TRUE(),
    [Active Customers vs Target %] >= 1.00, " ,
    [Active Customers vs Target %] >= 0.98, " ,
    "  "
)

```

New Customers =

```

SUMX(
    SUMMARIZE(
        Fact_Performance,
        Fact_Performance[Date],
        Fact_Performance[RegionKey],
        "RegionMonthNewCustomers",

```



```

        MAX(Fact_Performance[New_Customers])
    ),
    [RegionMonthNewCustomers]
)

```

```

Churned Customers =
SUMX(
    SUMMARIZE(
        Fact_Performance,
        Fact_Performance[Date],
        Fact_Performance[RegionKey],
        "RegionMonthChurnedCustomers",
        MAX(Fact_Performance[Churned_Customers])
    ),
    [RegionMonthChurnedCustomers]
)

```

```

Net New Customers =
[New Customers] - [Churned Customers]

```

```

Churned Customers Current Month =
CALCULATE(
    [Churned Customers],
    LASTDATE(Dim_Date[Date])
)

```

```

Churn Rate % =
DIVIDE(
    [Churned Customers Current Month],
    [Active Customers Closing]
)

```

```

Churn Rate RAG =
SWITCH(
    TRUE(),
    [Churn Rate %] <= 0.015, " ● ",
    [Churn Rate %] <= 0.0225, " ● ",
    " ● "
)

```

)

Total Complaints =

```
SUMX(
    SUMMARIZE(
        Fact_Performance,
        Fact_Performance[Date],
        Fact_Performance[RegionKey],
        "RegionMonthComplaints",
        MAX(Fact_Performance[Complaints])
    ),
    [RegionMonthComplaints]
)
```

Complaints Current Month =

```
CALCULATE(
    [Total Complaints],
    LASTDATE(Dim_Date[Date])
)
```

Complaint Rate =

```
DIVIDE(
    [Complaints Current Month],
    [Active Customers Closing]
) * 1000
```

Complaint Rate RAG =

```
SWITCH(
    TRUE(),
    [Complaint Rate] <= 2.0, " ,
    [Complaint Rate] <= 3.0, " ,
    "  "
)
```

OPERATIONS PILLAR

Total Transactions = SUM(Fact_Performance[Transaction_Count])

Transaction Target = SUM(Fact_Performance[Transaction_Target])

Transactions vs Target % = DIVIDE([Total Transactions], [Transaction Target])

Transactions RAG =

```
SWITCH(TRUE(),  
    [Transactions vs Target %] >= 1.00, " ● ",  
    [Transactions vs Target %] >= 0.95, " ● ",  
    " ● "  
)
```

Digital Transactions = SUM(Fact_Performance[Digital_Transaction_Count])

Digital Adoption % = DIVIDE([Digital Transactions], [Total Transactions])

Digital Adoption RAG =

```
SWITCH(TRUE(),  
    [Digital Adoption %] >= 0.78, " ● ",  
    [Digital Adoption %] >= 0.70, " ● ",  
    " ● "  
)
```

Total Tickets =

```
SUMX(  
    SUMMARIZE(  
        Fact_Performance,  
        Fact_Performance[Date],  
        Fact_Performance[RegionKey],  
        "RegionMonthTotalTickets",  
        MAX(Fact_Performance[Total_Tickets])  
    ),
```

```
[RegionMonthTotalTickets]
)
```

Tickets Within SLA =

```
SUMX(
  SUMMARIZE(
    Fact_Performance,
    Fact_Performance[Date],
    Fact_Performance[RegionKey],
    "RegionMonthTicketsWithinSLA",
    MAX(Fact_Performance[Tickets_Within_SLA])
  ),
  [RegionMonthTicketsWithinSLA]
)
```

SLA Compliance % = DIVIDE([Tickets Within SLA], [Total Tickets])

SLA RAG =

```
SWITCH(TRUE(),
  [SLA Compliance %] >= 0.95, " ,
  [SLA Compliance %] >= 0.90, " ,
  " "
)
```

Gross Loan Balance = SUM(Fact_Performance[Gross_Loan_Balance_EUR])

NPL Balance = SUM(Fact_Performance[NPL_Balance_EUR])

NPL Ratio % = DIVIDE([NPL Balance], [Gross Loan Balance])

NPL RAG =

```
SWITCH(TRUE(),
  [NPL Ratio %] <= 0.018, " ,
  [NPL Ratio %] <= 0.025, " ,
  " "
)
```

)

KPI framework note: Revenue Growth YoY remains part of the KPI framework because it is a standard executive metric. However, it is not displayed on the Executive Summary page because the workbook contains only 2025 data and therefore cannot produce populated SAMEPERIODLASTYEAR comparisons.

Dashboard Scope and Validation

Report Pages

- **Executive Summary:** KPI cards grouped into Revenue, Cost, Customer and Operations pillars. Revenue Growth YoY was omitted from the published dashboard because prior-year data is unavailable. Revenue per FTE appears alongside Revenue metrics due to layout considerations, although it is classified as an efficiency metric under the Cost pillar in the KPI framework.
- **Revenue Drill-Through:** revenue trend, budget comparison, regional contribution, and product contribution.
- **Operations Deep Dive:** SLA compliance, complaint rate, digital adoption, transactions, and NPL analysis.

Implementation Notes

- Cost vs Budget % was added to the Executive Summary as a management-control metric because it provides a more meaningful budget-performance view than displaying absolute operating cost alone.
- Map visuals were omitted because they are disabled by the tenant administrator; a country bar chart provides the same comparison more clearly.
- Technical key columns may remain visible during development and can be hidden before final publication without affecting relationships.
- Snapshot KPIs such as AUM and Active Customers use closing-balance logic rather than summing across months.
- RAG conditional formatting uses numeric status measures: 1 = Green, 2 = Amber, and 3 = Red.

Validation Checklist

- All KPI cards return values, except prior-year measures where 2024 data is unavailable.
- Slicers filter all intended visuals.
- Relationships are one-to-many and single-direction.

- Monthly revenue, operating cost, customer, AUM, transaction, SLA, and NPL results reconcile to the source workbook.
- Drill-through navigation and return behaviour are tested.
- During validation, several customer and operations KPIs were identified as semi-additive because the fact table is stored at Month × Region × Product grain while some operational fields are recorded at Month × Region grain. Measures such as Active Customers, New Customers, Churned Customers, Complaints, Tickets, Tickets Within SLA, and FTE were duplicated across four product rows. The issue was corrected in DAX by summarising values at Date × Region level and taking one value per region-month before aggregation. Snapshot KPIs such as Active Customers and FTE were evaluated using closing-month logic.
- Churn Rate, Complaint Rate, Active Customers, Net New Customers and Revenue per FTE were reviewed during validation. Initial calculations were distorted by duplicated Month × Region operational metrics across Product rows. Measures were revised to use region-level aggregation and closing-balance logic where appropriate.
- Active Customers, New Customers, Churned Customers, Churn Rate, Complaints, Complaint Rate, FTE, Revenue per FTE, Total Tickets, and Tickets Within SLA were revised after testing identified mixed-grain duplication across Product rows. Measures were redesigned using Date × Region aggregation logic and closing-month calculations where appropriate.

Business narrative: Germany and Luxembourg outperform throughout the year. France shows gradual improvement following mid-year operational pressure. Belgium experiences modest revenue decline while maintaining comparatively stable operational performance.

Portfolio Summary

Designed and built a company-wide Power BI KPI scorecard for a fictional Luxembourg financial services group. The solution uses a star schema, Power Query transformations, DAX measures, time-intelligence patterns, RAG conditional formatting, interactive slicers, drill-through analysis, and an operations deep dive. The dashboard supports management review across Revenue, Cost, Customer, and Operations.

